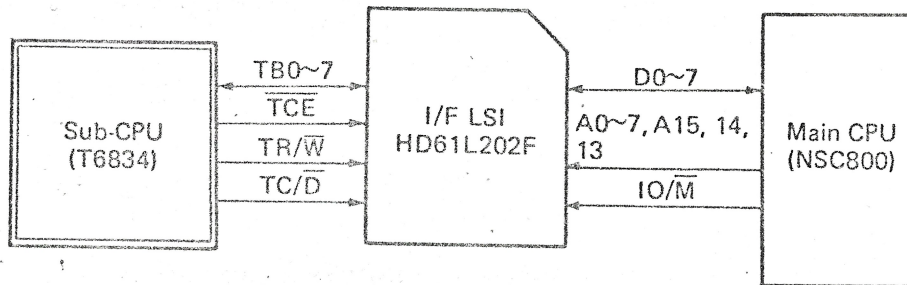


20. SUB CPU T6834

OUTLINE T6834 SUB-CPU

The T6834 sub-CPU mounted on the CCU board operated together with main CPU NSC800 via interface LSI Chip HD61L202F. The sub-CPU includes a mask ROM for controlling the sub-CPU operations.



20-1. DATA TRANSMISSION AND RECEPTION

The sub-CPU sends the chip enable signal (TCE), read/write signal (TR/W), and control/data signal (TC/D) to the HD61L202F. Data and commands are communicated via the data bus (TB0 to TB7) according to combinations of the content (0 or 1) of these signals.

TCE	TC/D	TR/W	TB0 to 7
1	X	X	High-impedance state
0	1	1	Reads data from the HD61L202F control register.
0	1	0	Writes data in the HD61L202F control register.
0	0	1	Reads data from HD61L202F data register XBTR.
0	0	0	Writes data in HD61L202F data register XBRR.

- 1) The sub-CPU reads the following data from the HD61L202F control register when $\overline{TCE} = 0$, $TC/\overline{D} = 1$, and $TR/\overline{W} = 1$:

D7	D6	D5	D4	D3	D2	D1	D0
OFFRQ	LBAT					XTRE	

OFFRQ: Set by and passed from the main CPU to the sub-CPU to request to switch the power off. When set to 1, the sub-CPU sets signal on A15 to high level, thereby switching VCC2 off.

LBAT: Set to 1 when the battery voltage is lowered. The sub-CPU reads this bit at 4-minute intervals. If it is 1, LOW BATT. is displayed.

XTRE: Indicates that the data buffer for transmitting data to the sub-CPU is empty.

- 2) The sub-CPU writes the following data in the HD61L202F control register when $\overline{TCE} = 0$, $TC/\overline{D} = 1$, $TR/\overline{W} = 0$:

D7	D6	D5	D4	D3	D2	D1	D0
TRQ1	TRQ0						

TRQ0: Set to 1 when the BREAK key is pressed. Causes an interrupt to the main CPU.

TRQ1: Resets the HD61L202F power save FF. When set to 1, the main CPU cannot enter the power save mode.

- 3) The sub-CPU reads the following data from the HD61L202F data register when $\overline{TCE} = 0$, $TC/\overline{D} = 0$, and $TR/\overline{W} = 1$:

D7	D6	D5	D4	D3	D2	D1	D0
			XBTR				
Data to be transmitted from the main CPU to the sub-CPU							

- 4) The sub-CPU writes the following data in the HD61L202F data register when $\overline{TCE} = 0$, $TC/\overline{D} = 0$, and $TR/\overline{W} = 0$:

D7	D6	D6	D5	D4	D3	D2	D1	D0
				XBRR				
Data to be transmitted from the sub-CPU to the main CPU								

- 5) Commands to be transmitted from sub-CPU to main CPU

Data	Command	Explanation
01	Shift On	Shift key detection
02	Shift Off	Shift key detection
03	Time Up	Timer time-up
04	Request Power Off	Off key detection
05	Break (On)	On key detection
06	Low Battery	Battery voltage is lowered.
07	Request Timer Off	Auto power off
08	Control-S	CTRL and S key detection

6) Commands to be transmitted from main CPU to sub-CPU

Command	Number of bytes (command and operand)	Explanation
DUMMY	1	No operation
Time Call	1	Reads year, month, day, day of the week, hour, minute, and second data items.
Stick	1	Reads the cursor key.
Strig	1	Checks user defined keys F6 and F12.
Strig 1	1	Checks space assignment.
Ram Read	3	Reads data from CCU board RAM.
Ram Write	4	Writes data in CCU board RAM.
Scroll Set	3	Specifies the starting and end lines for scrolling.
Scroll Exec	1	Scroll execution
Line Clear	2	Clears specified line display.
Time Set	9	Sets year, month, day, day of the week, hour, minute, and second data items.
Others (60 commands)		

20-2. RAM (2k bytes)

Addresses of the 2k byte RAM mounted on the CGU board are specified by signals A0 to A11 from the sub-CPU. (This is controlled by the sub-CPU and is separated from the RAM of the main CPU.)

The following items are written in the RAM:

- Key buffer data: 256 bytes (for 127 keys) *CC00 - CCFF*
- User defined characters: 512 bytes *CA00 - CBFF*
- User defined keys: 512 bytes *C800 - C9FF*
- Start program: 512 bytes *CE00 - CFFF*
- Unused area: 256 bytes *?*
- *System Data:* *CD00 - CDFF*

20-3. DEFAULT ITEMS

When the ALL RESET switch is pressed, the following conditions are set:

- The display screen is cleared.
- The cursor is positioned to (0, 0). (When the POWER switch is set to ON, before message "Copyright——" is displayed.)
- Key buffer is cleared.
- The user defined key and character areas are initialized.
- The time data is set as: 1 January 2001 Saturday 0 hour 0 minute 0 second.
- The alarm is cleared.

20-4. CLOCK FUNCTION

Since the sub-CPU has a clock function, VCCI is always supplied, even if the display is switched off by pressing the **OFF** key; therefore, the clock function is not interrupted and pressing of the **ON** key can be detected at any time. The crystal oscillator supplies 32.768 kHz signals. Its error is ± 30 seconds per month. The time display is made in the 24-hour system and a 1-channel alarm is provided.

20-5. KEY MATRIX

SFT	CTR							▶ A1
HOME		GRP	NUM	OFF			ON	BZ
[⁽³⁾	; +	: *	] }	9)	0 !	- =	^ ~	S8
. °	/ . ?	? -	CR	0 ⁽⁶⁾	P	@ ,	[{	S7
F1	F2	F3	F4	F5	F6			S6
1 !	2 "	3 #	4 \$	5 %	6 &	7 '	8 (	S5
Q	W	E	R	T	Y	U ⁽⁴⁾	I ⁽⁵⁾	S4
A	S	D	F	G	H	J ⁽¹⁾	K ⁽²⁾	S3
Z	X	C	V	B	N	M ⁽⁰⁾	.	S2
INS	DEL	→	←	↑	↓	SPC		S1
K1	K2	K3	K4	K5	K6	K7	K8	

20-6. KEY CODES (ASCII)

	0	1	2	3	4	5	6	7	8	9		
0												
1			SP	0	@	P	,	P	/	2	1	ε
2			"	1	A	Q	a	q	:	3	0	σ
3			#	2	B	R	b	r	;	4	8	θ
4				3	C	S	c	s	?	5	X	
5			\$	4	D	T	d	t	!	U	λ	
6			%	5	E	U	e	u	]	V	μ	
7	BEL		&	6	F	V	f	v		Y	ρ	
8	BS		'	7	G	W	g	w		À	π	
9			(	8	H	X	h	x	A	á	τ	
A	LF		)	9	I	Y	i	y	·	â	Ø	
B	HOM		*	:	J	Z	j	z	0	ä	x	
C	CLS	→	+	:	K	[	k	{	Q	å	ω	
D	CR	←	,	<	L	¥	l			ä	ν	
E		↖	-	=	M	]	m	}		ä	β	
F		↗	.	>	N	^	n	~		ï	γ	
			/	?	O	-	o	¿		í	δ	÷

NUM Mode